

21 Summarising data

Introductory problem

The magnetic dipole of an electron is measured in a very sensitive experiment 3 times. The values are 2.000 001 5, 2.000 000 12 and 2.000 000 9. Does this support the theory that the magnetic dipole is 2?

Huge amounts of data are collected in scientific experiments and social surveys. Data can be very difficult to interpret, so we often summarise the important features using statistics – numbers or diagrams which represent something about the dataset, such as where its centre lies or how spread out it is.

21A Some important concepts in statistics

Perhaps the most important concept in statistics is the difference between a sample and the population.

The **population** is the entire group which we are interested in. This might be everyone in the world, or just all 18-year-old girls in a school.

The **sample** is a group of data collected from the population. It is often assumed that any statistics of the sample reflect the statistics of the population. This may not be the case for three reasons:

1. The sample may be unrepresentative of the population. If you want to know the average height of boys in a school and you only ask the basketball team, you may get an overestimate.
2. Even with a good sample, it is unlikely you will get exactly the same statistic as for the whole population.

In this chapter you will learn:

- about the difference between a sample and the entire set of data
- about different types of data
- different ways to measure how spread out the data is
- how to work with data that has been summarised



Although statistics are often helpful, there are many ways in which they can be misleading.

Quantum theory, an area of physics, suggests that at the atomic level all physical variables are actually discrete! Surprisingly, evidence to support this can be found by looking very carefully at the colour of objects radiating heat at different temperatures. If you want to know more about this, try researching the ultra-violet catastrophe.



3. Perhaps the most subtle reason is the fact that for some statistics we can show mathematically that the statistic for the sample will systematically tend to be different from the statistic of the population.

To avoid the first of these issues it is generally good practice to collect a **random sample**. This means that each member of the population is equally likely to be selected for the sample and the probability of selecting any member of the population is independent of the selection of any other member. This can be very difficult to achieve.

The other two issues are more difficult to avoid. In particular, even with a random sample, it is possible to get a correct but unusual data value. Such a value is called an **outlier**, and is different from an **anomaly**, which refers to a measurement error.

Another important distinction is between different types of numerical data which may be collected:

Discrete data is data that takes only particular, predefined values. Shoe size is an example of discrete data. Data does not have to take only integer values to be discrete.

Continuous data is data which can take any value in a given range. Foot length is an example of continuous data. To work with continuous data we need to group it together, so we would not count how many people have a foot length of 20.034 621... cm; we might instead count how many people have a foot length between 20.0 and 20.1 cm.

Exercise 21A

1. Decide which of these variables are continuous and which are discrete:
 - (a) days in a month
 - (b) shoe size
 - (c) length of foot
 - (d) weight of a gerbil
 - (e) length of arm to the nearest centimetre
 - (f) age in completed years.
2. For each of the following suggest why the sample taken is not random:
 - (a) a questionnaire to find out attitudes towards a new shopping development taken on the high street at 4 p.m. on a Saturday

- (b) interviewing students in a lesson about truancy rates
- (c) an internet survey of voting intentions
- (d) a cholesterol test from one person in each household in a street.

21B Measures of spread

Many statistical calculations, such as the mean, median and mode, focus on finding a measure of the centre of the data. Others, such as the range and the interquartile range, give you a measure of how spread out the data is. The problem with these two measures is that they do not take into account all of the data.

Section Y of Prior Learning reminds

you how to calculate each of these statistics.



What are the pros and cons of representing a complicated set of data by a single number?



In everyday language the word average is often used without specifying whether it refers to the mean, the mode or the median. Potentially this can lead to some confusing newspaper headlines. In mathematics it is important to be precise.

A commonly used measure that does use all the data is called the **standard deviation**, and for a sample it is given the symbol s_n . This measures the average distance of data items from the mean.

KEY POINT 21.1

Standard deviation formula

$$s_n^2 = \sum_i \frac{(x_i - \bar{x})^2}{n}$$

Where x_i is the i th data value, $\bar{x}$ is the mean of all the data values and n is the number of data items.

Although this is the defining formula for the standard deviation, it is not normally the formula used to calculate it. Fill-in proof 24 'An alternative formula for variance' shows that an alternative and much more useful formula is:

KEY POINT 21.2

$$s_n^2 = \overline{x^2} - \bar{x}^2$$



The word *range* has several different meanings in mathematics, both technical and informal. In particular, the range of a function (chapter 5) means the set of values that the function can take; but the range of data is a *single number* representing the length of the interval in which the data lies.

EXAM HINT

$\overline{x^2}$ is the average of data items after they have been squared.
 $\overline{x}^2$ is the square of the average of the original data items.
These two things are not the same! In fact, the first one is always larger, so s_n^2 cannot be negative.

It may help you to remember this formula as 'the mean of the squares minus the square of the mean'.

The square of the standard deviation is called the **variance**. It does not give any direct measure of spread, but if you study the Statistics option (Topic option 7) you will see that it is, algebraically, very convenient to work with.

EXAM HINT

Make sure you know whether a question is asking for standard deviation or variance.

Worked example 21.1

Find the range, interquartile range and standard deviation of the following numbers:

1, 12, 9, 9, 15, 7, 5

Order the data then split into two halves

Find $\overline{x}$ and $\overline{x^2}$

$$\text{range} = 15 - 1 = 14$$

Data in order: 1, 5, 7, 9, 9, 12, 15

LQ = 5, UQ = 12 so IQR = 7

	x	x^2
	1	1
	5	25
	7	49
	9	81
	9	81
	12	144
	15	225
Mean:	8.29	86.57

$$s_n = \sqrt{86.57 - 8.29^2}$$

$$= \sqrt{17.92}$$

$$\approx 4.23 \text{ (3SF)}$$

EXAM HINT

Most graphical calculators have methods for finding all these statistics. See Calculator skills sheets 11 and 12. However, you can still be asked to do it without a calculator.



As a rough guide, about two thirds of the data will be less than one standard deviation away from the mean. In the above example two data items (1 and 15) out of seven are more than 4.23 away from the mean (8.29). This can be a useful quick check. In a large data set nearly all the data will be within two standard deviations from the mean, and anything more than three standard deviations from the mean is very unusual.

Sometimes you may have to work with summary statistics concerning the data, but do not have the original data. In this situation, it is often best to work with the sum of all the data values and the sum of the all the squared data values.

Worked example 21.2

Six trophies have a mean height of 28 cm and a standard deviation of 6 cm. A seventh trophy is added with height 38 cm. Find the mean and standard deviation of the seven trophies.

Find the total height of the first six trophies. This is because a total is very easy to combine with a new piece of data

$$\Sigma x = 28 \times 6 = 168$$

Use the formula for variance to find the total of the squares of the heights of the original six trophies

$$6^2 = \frac{(\Sigma x^2)}{6} - 28^2$$

$$\Sigma x^2 = 4920$$

Find the new totals

$$\Sigma x = 168 + 38 = 206$$

$$\Sigma x^2 = 4920 + 38^2 = 6364$$

Find the new mean

$$\bar{x} = \frac{206}{7} = 29.4 \text{ (3SF)}$$

Find the new variance

$$s_n^2 = \frac{6364}{7} - 29.4^2 = 43.1 \text{ (3SF)}$$

$$s_n = 6.57 \text{ (3SF)}$$

Exercise 21B



1. For each set of data below calculate the standard deviation:

- (a) (i) 19.0, 23.4, 36.2, 18.7, 15.7
- (ii) 0.4, -1.3, 7.9, 8.4, -9.4
- (b) (i) 28, 31, 54, 28, 17, 30
- (ii) 60, 18, 42, 113, 95, 23



2. For each set of data below calculate the standard deviation:

- (a) 1, 1, 2, 3, 5
- (b) 3, -2, 4, -2, 5, 2

3. The ordered set of data 5, 5, 7, 8, 9, x , 13 has interquartile range equal to 7.

- (a) Find the value of x .
- (b) Find the standard deviation of the data set. [5 marks]

4. Consider the five numbers, 2, 5, 9, x and y . The mean of the numbers is 5 and the variance is 6. Find the value of xy . [7 marks]

5. In five tests Suewan has an average of 23 marks and a standard deviation of 4 marks. In her sixth test she scores 32. What is the overall standard deviation of her marks? [7 marks]

6. The mean of a set of 15 data items is 600 and the standard deviation is 12. Another piece of data is discovered and the new mean is 600.25. What is the new standard deviation? [7 marks]

7. If the sum of 20 pieces of data is 1542 find the smallest possible value of $\sum x_i^2$. [4 marks]

8. (a) Explain why for any set of data $x_i - \bar{x}$ is no greater than the range.

(b) By considering the formula $s_n = \sqrt{\sum_i \frac{(x_i - \bar{x})^2}{n}}$, prove that the standard deviation is always less than or equal to the range. [4 marks]

21C Frequency tables and grouped data

It is common to summarise a large quantity of data in a **frequency distribution** table. This is a list of all the values the data takes, along with how often they occur. We could convert this into a list of all the data values and calculate the statistics as we had before, but it is enough to just imagine writing out a list, 16 ones, two twos, etc.

Worked example 21.3

Find the mean number of passengers observed in cars as they passed a school.

Passengers	Frequency
0	32
1	16
2	2
3 or more	0

(none in the first group, 16 in the second group 2, and 4 in the third group)

$$\begin{aligned}\text{total number of passengers} &= (32 \times 0) + (16 \times 1) + (2 \times 2) + 0 = 20 \\ \text{mean} &= \frac{20}{50} = 0.4\end{aligned}$$

EXAM HINT

The mean does not have to be an achievable value; do not round to the nearest whole number.

This method suggests an important formula.

KEY POINT 21.3

Finding the mean from a frequency table:

$$\bar{x} = \frac{\sum x_i f_i}{n}$$

where f_i is the frequency of the i th data value and

$n = \sum_i f_i$ is the total number of data items.

We can work out $\overline{x^2}$ in a similar way, which gives the following formula for standard deviation.

KEY POINT 21.4

Standard deviation from a frequency table:

$$s_n^2 = \frac{\sum_i x_i^2 f_i}{n} - \bar{x}^2$$

EXAM HINT

The notation $[a, c[$ means the same as $a < x < c$. You might also see this written as $[a, c)$.

In Worked example 21.3, we knew the exact data values, but when we are dealing with grouped data, we do not have this level of precision. In order to work out the mean and standard deviation, our best and simplest assumption is that all the original values in a particular group are located at the centre of the group, called the mid-interval value. To find the centre of the group we take the mean of the largest and the smallest possible values in the group, called the upper and lower interval boundaries.

Worked example 21.4

Find the mean and standard deviation of the weight of eggs produced by a chicken farm. Explain why these answers are only estimates.

Weight of eggs, in g	Frequency
$[100, 120[$	26
$[120, 140[$	52
$[140, 160[$	84
$[160, 180[$	60
$[180, 200[$	12

Make a table using the mid-interval value for each group

x_i	f_i	$x_i f_i$	$x_i^2 f_i$
110	26	2860	314 600
130	52	6760	878 800
150	84	12 600	1 890 000
170	60	10 200	1 734 000
190	12	2280	433 200
Sum:	234	34 700	5 250 600

Apply the formulae

EXAM HINT

Whenever you find a mean or a standard deviation it is always worth checking that the numbers make sense in context. Given the data, an average of about 150 g here seems reasonable.

$$\bar{x} = \frac{\sum x_i f_i}{n} = \frac{34700}{234} = 148.3 \text{ g (3SF)}$$

$$s_n^2 = \frac{\sum x_i^2 f_i}{n} - \bar{x}^2 = \frac{5250600}{234} - 148.3^2 = 448.4$$

$$\text{Therefore } s_n = 21.2 \text{ g (3SF)}$$

These answers are only estimates because we have assumed that all the values in each group are at the centre, rather than using the actual data.

Sometimes the endpoints of the intervals shown in the table are not the actual smallest and largest possible values in that group. For example, when measuring length in centimetres it is common to round the values to the nearest integer, so 10–15 actually means [9.5, 15.5]. To find the mid-interval values we must first identify the actual interval boundaries.

Worked example 21.5

Estimate the mean of this data:

Age	Frequency
10 to 12	27
13 to 15	44
16 to 19	29

Carefully decide on the upper and lower interval boundaries. There should be no 'gaps' between the groups, because age is continuous data. You are 12 years old until your 13th birthday

Group	x_i	f_i	$x_i f_i$
[10, 13[	11.5	27	310.5
[13, 16[	14.5	44	638
[16, 20[	18	29	522
Sum:		100	1470.5

$$\bar{x} = \frac{\sum x_i f_i}{n} = \frac{1470.5}{100} = 14.7 \text{ (3SF)}$$

Exercise 21C

1. Calculate the mean and standard deviation of each data set:

(a)

x	Frequency
0	16
1	22
2	8
3	4
4	0

(b)

x	Frequency
-1	10
0	8
1	5
2	1
3	1



2. Calculate the mean and standard deviation for each data set:

(a)

x	Frequency
10	7
12	19
14	2
16	0
18	2

(b)

x	Frequency
0.1	16
0.2	15
0.3	12
0.4	9
0.5	8

3. A group is described as '17 – 20'. State the upper and lower boundaries of this group if it is measuring:
- age in completed years
 - number of pencils
 - length of a worm to the nearest centimetre
 - hourly earnings, rounded up to whole dollars.



4. Find the mean and standard deviation of each of the following sets of data:

(a) (i) x is the time taken to complete a puzzle in seconds

x	Frequency
[0,15[	19
[15,30[	15
[30,45[	7
[45,60[	5
[60,90[	4

(ii) x is the weight of plants in grams

x	Frequency
[50,100[	17
[100,200[	23
[200,300[	42
[300,500[	21
[500,1000[	5

- (b) (i) x is the length of fossils found in a geological dig, to the nearest centimetre

x	Frequency
0 to 4	71
5 to 10	43
11 to 15	22
16 to 30	6

- (ii) x is the power consumption of light bulbs, to the nearest watt

x	Frequency
90 to 95	17
96 to 100	23
101 to 105	42
106 to 110	21
111 to 120	5

- (c) (i) x is the age of children in a hospital ward

x	Frequency
0 to 2	12
3 to 5	15
6 to 10	7
11 to 16	6
17 to 18	3

- (ii) x is the amount of tips paid in a restaurant, rounded down to the nearest dollar

x	Frequency
0 to 5	17
6 to 10	29
11 to 20	44
21 to 30	16
31 to 50	8



5. In a sample of 50 boxes of eggs, the number of broken eggs per box is shown below:

Number of broken eggs	0	1	2	3	4	5	6
Number of boxes	17	8	7	7	6	5	0

- (a) Calculate the median number of broken eggs per box.
 (b) Calculate the mean number of broken eggs per box. [4 marks]
6. The mean of the data in the table is 32 and the variance is 136. Find the possible values of p and q .

x	Frequency
20	12
40	q
p	8

[8 marks]

Summary

- Most of statistics is based on trying to infer properties of a **population** based upon a **sample** from that population.
- To get a representative sample of the population, it is good practice to collect a **random sample**, where each member of the population is equally likely to be selected for the sample and the probability of selecting a member of the population is independent.
- An **outlier** is a correct but unusual data value.
- An **anomaly** is an unusual data value caused by a measurement error.
- Discrete data** takes only a predefined value (it does not have to be an integer!).
- Continuous data** can take any value in a given range. This type of data is generally grouped before we can work with it.
- Standard deviation** (s_n) is a measure of how spread out the data is relative to the data's mean, and it takes into account all of the data.
- The square of the standard deviation is called the **variance** and it has the formula:

$$s_n^2 = \frac{\sum (x_i - \bar{x})^2}{n}$$
 or more commonly: $s_n^2 = \overline{x^2} - \bar{x}^2$
- Large datasets are summarised in **frequency distribution** tables and the mean and standard deviations can be calculated from these tables using the formulae:

$$\bar{x} = \frac{\sum x_i f_i}{n}$$

where f_i is the frequency of the i th data value and $n = \sum f_i$ is the total number of data items,

and:

$$s_n^2 = \frac{\sum_i x_i^2 f_i}{n} - \bar{x}^2$$

- When investigating grouped data we must assume that every element has the mid-interval value of the group (the mean of the upper and lower boundaries).
- The methods for calculating statistics for grouped data vary slightly depending upon whether the data is discrete or continuous.

Introductory problem revisited

The magnetic dipole of an electron is measured in a very sensitive experiment 3 times. The values are 2.000 000 15, 2.000 000 12 and 2.000 000 9. Does this support the theory that the magnetic dipole is 2?

The average magnetic dipole is 2.000 000 12 which is pretty close to 2, but the standard deviation in the measurements is 0.000 000 245, and so the mean is approximately 5 sample standard deviations away from 2. Within the natural variation observed, the magnetic dipole cannot be said to be 2.



The difference between 2.000 001 2 and 2 might seem trivial, but it was this difference which inspired Richard Feynman to create a new theory of physics called Quantum electrodynamics, which did indeed predict this tiny difference from 2! This is an example of theory driving experiment which in turn creates new theory – the interplay between theoretical mathematics and reality.

Mixed examination practice 21

Short questions

1. A student takes the bus to school every morning. She records the length of the time, in minutes, she waits for the bus on 12 randomly chosen days. The data is summarised by:

$$\sum_{i=1}^{12} x_i = 49 \text{ and } \sum_{i=1}^{12} x_i^2 = 305.7$$

Calculate:

- (a) the mean time she spends waiting for the bus
(b) the standard deviation of the times.

[5 marks]

2. The average wavelength of light in nanometres emitted by a glowing wire is measured on 50 different occasions and the results are given below:

Wavelength in nm (λ)	Frequency
600–640	22
640–680	18
680–720	x
720–760	y

The mean of λ is calculated from this table as 653.6.

- (a) Find the values of x and y .
(b) Calculate an estimate of the variance.
(c) Explain why this is only an estimate.

[9 marks]

3. An experiment was conducted on the reaction times of 15 students (t) in seconds. The results were that the average reaction time was 0.2 s and the variance was 0.0025 s^2 . A 16th student is observed later. She has a reaction time of 0.16 s. Find the new mean and standard deviation.

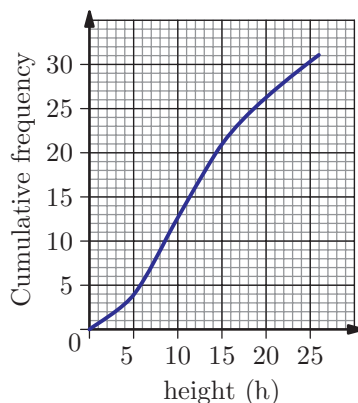
[8 marks]

4. The variance of two data items is k . Find an expression in terms of k for the range.

[5 marks]

Long questions

1. The following is the cumulative frequency diagram for the heights of 30 plants, given in centimetres.



- (a) Use the diagram to estimate the median height.
 (b) Complete the following frequency table:

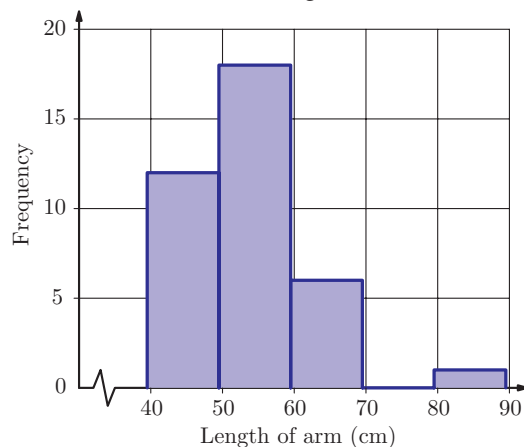
Height (h)	Frequency
$0 < h \leq 5$	4
$5 < h \leq 10$	9
$10 < h \leq 15$	
$15 < h \leq 20$	
$20 < h \leq 25$	

- (c) Hence estimate the mean height.

[8 marks]

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2. The following histogram shows the length of the arms of 37 children in a classroom in Lithuania, given to the nearest cm.



- (a) Explain why the bar representing the first group goes below 40.
 (b) Complete the following frequency distribution:

Length	Frequency
40–49	12
50–59	
60–69	
70–79	
80–89	

- (c) Use this data to estimate the mean and the standard deviation of the data.
 (d) Give one reason to explain why the average arm length of all children in Lithuania might be different from the value found above.
3. The frequency distribution of the number of cars in households on a street is given below:

Number of cars	Frequency
1	a
2	b

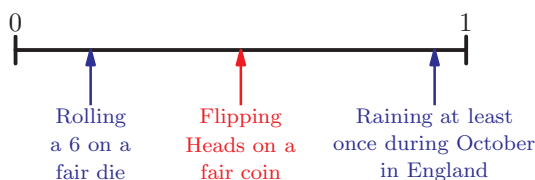
- (a) Find an expression for the mean of the number of cars and show that the variance is given by $\frac{ab}{(a+b)^2}$.
 (b) Prove that it is impossible for the mean to equal the variance.
 (c) If the number of households with one car is three times larger than the number of households with two cars find the mean and the standard deviation in the number of cars.

22 Probability

Introductory problem

A woman gives birth to non-identical twins. One of them is a girl. What is the probability that the other one is a girl?

In real life we often deal with uncertain events, but not all events are equally uncertain. It is not certain that the next Steven Spielberg film will be a big hit, nor is it certain that it will snow in India next summer. However, intuitively, these two events do not seem equally likely. We can put events on a scale with impossible at one end and certain at the other and assign a number between 0 and 1 to indicate where the event is on this line. This number is called the probability of the event.



For probability to be useful, it must be more than just a reflection of past experience. We also have to be able to predict probabilities of events in the future, and use these predictions to make decisions. This is the focus of this chapter.

22A Introduction to probability

We can estimate the probability of an event practically by doing an experiment repeatedly; this is known as empirical or experimental probability:

KEY POINT 22.1

The probability of an event 'A' occurring is denoted by $P(A)$. From observation, we can estimate $P(A)$ as:

$$P(A) = \frac{\text{Number of times } A \text{ occurs}}{\text{Number of times } A \text{ could have occurred}}$$

In this chapter you will learn:

- how probability can be estimated from data
- how probability can be predicted theoretically
- how to work out probabilities when you are interested in more than one outcome
- how to work out the probability of a sequence of events occurring
- how to work out probabilities of simple functions of independent random events
- how counting principles can be used to calculate probabilities
- how being given additional information changes our estimate of a probability
- how to infer information from changes in probabilities.



Probability is one of the most recent additions to the field of mathematics. It was formalised by the mathematician Pierre de Fermat (1601–1665) in response to a request from his patron, the Chevalier de Méré, a notorious gambler who wanted help at the gambling table.

Probability is revised
in Prior learning
section H on the
CD-ROM.



However, in some situations it is possible to predict the probability *before* the experiment (theoretical probability). To do this we need to be able to list all the possible outcomes. This list is called the **sample space**.

For example, when you toss a coin there are two possible outcomes: Heads or Tails. Since both are equally likely (for a 'fair' coin), the probability of each must be one half. This leads to a theoretical definition of probability:

KEY POINT 22.2

$$P(A) = \frac{\text{number of times } A \text{ occurs in the sample space}}{\text{number of items in the sample space}}$$

Again we have $0 \leq P(A) \leq 1$. We can now add an interpretation to $P(A)$. If $P(A) = 0$ then the event A is impossible. If $P(A) = 1$ then event A is certain. As $P(A)$ rises the likelihood of A occurring increases.

You might think it seems obvious that these two definitions are equivalent. However, it is quite tricky to prove. If you would like to see how it is done you might like to research the law of large numbers.



There are two possible outcomes when you enter a lottery. Either you win or you do not win, but this does not mean that the probability of winning is one half, since there is no reason to believe that both outcomes are equally likely. Many mistakes in probability come from this type of error.



Worked example 22.1

- For a family with two children, list the sample space for the sexes of the children, assuming no twins.
- Hence find the theoretical probability that the two children are a boy and a girl.

A table provides a systematic way to list the possibilities

There are four outcomes, and two of them are a boy and a girl

(a)

First child	Second child
Boy	Boy
Boy	Girl
Girl	Boy
Girl	Girl

(b) $P(\text{a boy and a girl}) = \frac{2}{4} = \frac{1}{2}$

EXAM HINT

Although the 'boy first, girl second' and 'girl first, boy second' cases can both be described as 'a boy and a girl', we must count them separately in the sample space.

EXAM HINT

'Die' is the singular of 'dice'.

Worked example 22.2

What is the probability of getting a prime number when you roll a six-sided die?

We can list all the possible outcomes and they are all equally likely

Possible outcomes are 1, 2, 3, 4, 5, 6

We can identify how many of them are prime

2, 3 and 5 are prime, which is 3 outcomes out of a sample space of 6.

We can write this as a probability

So probability is $\frac{3}{6} = \frac{1}{2}$

An event either happens or it does not. Everything other than the event happening is called the **complement** of the event. For example, the complement of rolling a 6 on a die is rolling 'not a 6' so 1, 2, 3, 4 or 5. The complement of A is given the symbol A' .

An event and its complement are mutually exclusive (they can't both happen), but we know that one of the two must happen so they have a total probability of 1. We can therefore deduce a formula linking the probability of an event and its complement.

KEY POINT 22.3

$$P(A) + P(A') = 1$$

Suppose that in a board game, players face a penalty whenever the sum of two rolled dice is 7. We do not care how the 7 is achieved – it may be from a 1 and a 6, or a 3 and a 4. In such a situation it is useful to use a probability grid diagram which lists the sample space with each possibility of the first event on one axis, each possibility of the second event on the other axis, giving all possible results in the cells of the table.



Saying that an event either happens or it does not is called the law of the excluded middle, and it is a basic axiom of standard logic. However, there is an alternative logical system called fuzzy logic where an event could be in a state of 'maybe happening'. As well as lots of real world applications there is a philosophical physics problem called Schrödinger's Cat which applies this idea.

Worked example 22.3

What is the probability of getting a sum of 7 when two dice are rolled?

Draw a probability grid diagram showing all possible totals when two dice are rolled

		Die A					
Die B		1	2	3	4	5	6
	1	2	3	4	5	6	7
	2	3	4	5	6	7	8
	3	4	5	6	7	8	9
	4	5	6	7	8	9	10
	5	6	7	8	9	10	11
	6	7	8	9	10	11	12

Count how many items are in the sample space

36 items in the sample space, each of equal probability.

Count how many sums of 7 there are

6 ways to get 7 therefore the probability of a sum of 7 is $\frac{6}{36} = \frac{1}{6}$

EXAM HINT

Notice in the grid in Worked example 22.3 that a '3' on die A and a '1' on die B has to be counted as a separate event from a '1' on die A and '3' on die B. However, a '3' on die A and a '3' on die B is only one event. This often causes confusion.

We are often interested in more complicated events, such as 'getting a grade A in Extended Essay or TOK'.

In everyday language the word 'or' can be ambiguous. If you say 'Everyone at the party is a doctor or a lawyer' you generally do not mean that each person could be both. However, in a game, if you say 'I win if I get a black number or an even number' you would also expect to win if you got a black even number. In probability we use the word 'or' in the second sense: ' A or B ' means A or B or both could happen.

Worked example 22.4

When a die is rolled find the probability that the outcome is odd or a prime number.

List the sample space

Possible outcomes: 1, 2, 3, 4, 5, 6

List the events which satisfy the condition

Odd or prime: 1, 2, 3, 5

$$\text{Probability} = \frac{4}{6} = \frac{2}{3}$$

Exercise 22A

- List the sample space for each of the following:
 - a fair 6 sided die
 - arrangements of the letters 'RED'
 - the sexes of a 3-child family
 - a six sided die with 3 sides labelled '1' and the remaining sides labelled 2, 3, 4.
- In a standard pack of 52 playing cards there are 4 different suits (red hearts, red diamonds, black clubs and black spades). In each suit there are number cards from 2 to 10, then four 'picture cards'; jack, queen, king and ace. Find the probability that a randomly chosen card is:
 - red
 - a spade
 - a jack
 - a picture card
 - a black number card
 - a club picture card
 - not a heart
 - not a picture card
 - a club or a picture card
 - a red card or a number card
 - a red number card strictly between three and nine
 - a picture card that is not a jack
- A bag contains three different kinds of marble: six are red, four are blue and five are yellow. One marble is taken from the bag. Calculate the probability that it is:
 - red
 - yellow
 - not blue
 - not red

EXAM HINT

You can give probability as a fraction, a decimal or as a percentage.

- (c) (i) blue or yellow (ii) red or blue
- (d) (i) green (ii) not green
- (e) (i) neither red nor yellow (ii) neither yellow nor blue
- (f) (i) red or green (ii) neither blue nor green
- (g) (i) red and green (ii) red and blue

4. By means of an example show that $P(A) + P(B) = 1$ does not mean that B is the complement of A .
5. Two fair six-sided dice numbered one to six are rolled. By drawing probability grid diagrams find the probability that:
- (a) the sum is 8
 - (b) the product is greater than or equal to 8
 - (c) the product is 24 or 12
 - (d) the largest value is 4
 - (e) the largest value is more than twice the smaller value
 - (f) the value on the first die divided by the value on the second die is a whole number.
6. A fair four-sided die (numbered 1 to 4) and a fair eight-sided die (numbered 1 to 8) are rolled. Find the probability that:
- (a) the sum is 8
 - (b) the product is greater than or equal to 8
 - (c) the product is 24 or 12
 - (d) the largest value is 4
 - (e) the largest value is more than twice the smaller value
 - (f) the value on the eight-sided die divided by the value on the four-sided die is a whole number.
7. Two fair six-sided dice are thrown and the score is the highest common factor of the two outcomes. If this were done 180 times how many times would you expect the score to be 1? [5 marks]
8. Three fair six-sided dice are thrown, and the score is the sum of the three results. What is the probability that the score is less than 6? [5 marks]

22B Combined events and Venn diagrams

In this section we shall generalise the sample space method so we can efficiently calculate probabilities when we are interested in more than one outcome.

Which is more likely when you roll a die once:

- getting a prime number *and* an odd number?
- getting a prime number *or* an odd number?

The first possibility is restrictive; we have to satisfy both conditions. The second has more possibilities – we can satisfy either condition. So the second must be more likely.

These are examples of two of the most common ways of combining events: Intersection (in normal language ‘and’) and Union (in normal language ‘or’). These are given the following symbols, which might be used in the examination, so get used to them!

$A \cap B$ is the **intersection** of A and B , meaning when both A and B happens.

$A \cup B$ is the **union** of A and B , meaning when either A happens, or B happens, or both happen.

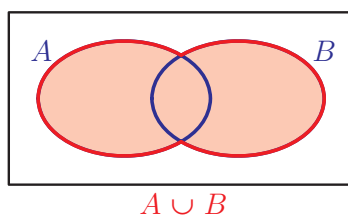
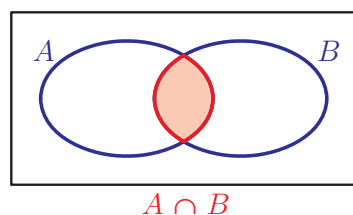


If we have neither apples nor pears then we have no apples and no pears. In set notation this can be written as $(A \cup B)' = A' \cap B'$. This is one of De Morgan's laws - a description of some of the algebraic rules obeyed by sets and hence probability. You will meet these again if you take the Discrete mathematics option (Topic option 10) or the Sets, relations and groups option (Topic option 8).



Why do mathematicians not just use simple words? One of the reasons for this is that everyday language can be ambiguous. If I say that I play rugby or hockey some people may think this means I do not play both. Mathematicians hate ambiguity.

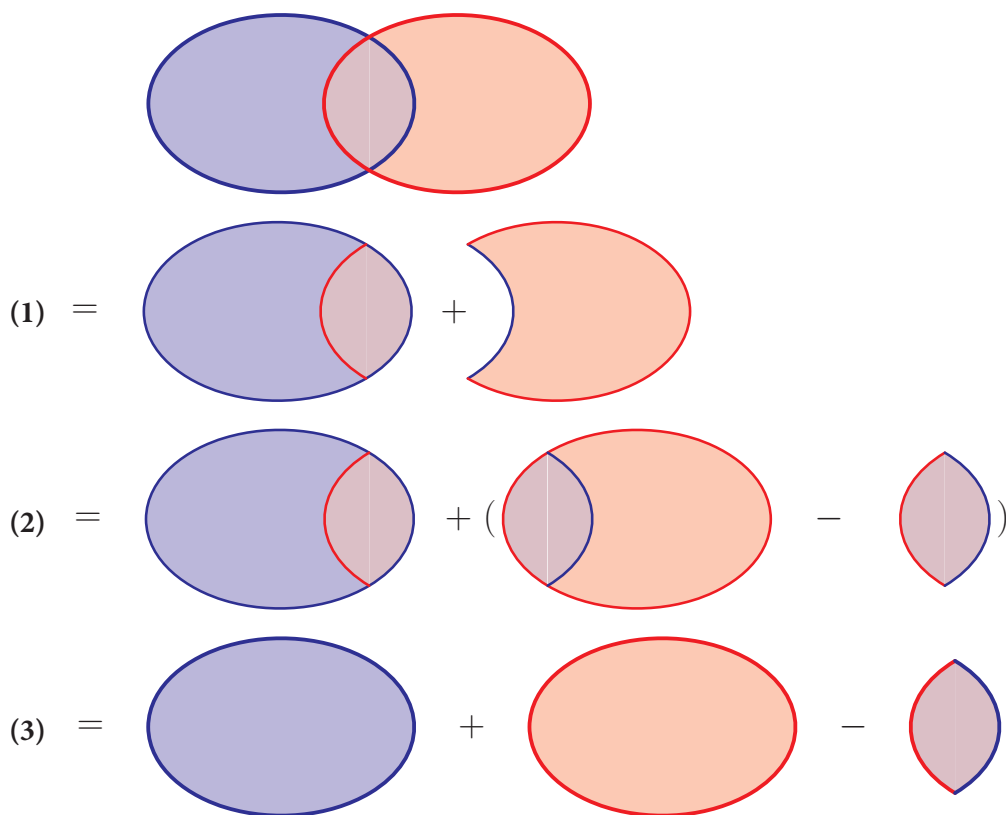
We can use **Venn diagrams** to illustrate the concepts of union and intersection:



See Prior learning J on the CD-ROM which explains the concept of Venn diagrams.



Venn diagrams can help us calculate the probability of the union of two events. In the following diagram, the region of the overlap can be seen to be composed of three sections, which can be broken down into their different regions.



This illustrates a very important formula.

KEY POINT 22.4

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

EXAM HINT

Be careful that you do not use the formula for mutually exclusive events unless you are sure that the events cannot both occur simultaneously.

We can interpret this formula as 'If you want to count the number of ways of getting A or B , count the number of ways of getting A and add to that the number of ways of getting B . However, we have then counted the number of ways of getting A and B twice, so we need to compensate by subtracting it'.

If there is no possibility of A and B occurring at the same time, then $P(A \cap B) = 0$. We call these events **mutually exclusive**, and the formula reduces to $P(A \cup B) = P(A) + P(B)$.

Worked example 22.5

A chocolate is selected randomly from a box. The probability of it containing nuts is $\frac{1}{4}$. The probability of it containing caramel is $\frac{1}{3}$. The probability of it containing both nuts and caramel is $\frac{1}{6}$. What is the probability of a randomly chosen chocolate containing either nuts or caramel or both?

Use the formula

$$\begin{aligned}P(\text{nuts} \cup \text{caramel}) &= P(\text{nuts}) + P(\text{caramel}) - P(\text{nuts} \cap \text{caramel}) \\&= \frac{1}{4} + \frac{1}{3} - \frac{1}{6} = \frac{5}{12}\end{aligned}$$

Exercise 22B

1. (a) (i) $P(A) = 0.4$, $P(B) = 0.3$ and $P(A \cap B) = 0.2$.
Find $P(A \cup B)$.
(ii) $P(A) = \frac{3}{10}$, $P(B) = \frac{4}{5}$ and $P(A \cap B) = \frac{1}{10}$.
Find $P(A \cup B)$.
- (b) (i) $P(A) = \frac{2}{3}$, $P(B) = \frac{1}{8}$ and $P(A \cup B) = \frac{5}{8}$.
Find $P(A \cap B)$.
(ii) $P(A) = 0.2$, $P(B) = 0.1$ and $P(A \cup B) = 0.25$.
Find $P(A \cap B)$.
- (c) (i) $P(A \cap B) = 20\%$, $P(A \cup B) = 0.4$ and $P(A) = \frac{1}{3}$.
Find $P(B)$.
(ii) $P(A \cup B) = 1$, $P(A \cap B) = 0$ and $P(B) = 0.8$.
Find $P(A)$.
- (d) (i) Find $P(A \cup B)$ if $P(A) = 0.4$, $P(B) = 0.3$ and A and B are mutually exclusive.
(ii) Find $P(A \cup B)$ if $P(A) = 0.1$, $P(B) = 0.01$ and A and B are mutually exclusive.

2. (a) (i) When a fruit pie is selected at random,

$$P(\text{it contains pears}) = \frac{1}{5} \text{ and } P(\text{it contains apples}) = \frac{1}{4}.$$

10% contain both apples and pears. What is the probability that it contains either apples or pears?

- (ii) In a library 80% of books are classed as fiction and 70% were written in the 20th century. Half of the books are 20th century fiction. What proportion of the books are either fiction or from the 20th century?
- (b) (i) 95% of pupils in a school play either football or tennis. The probability of a randomly chosen pupil playing football is $\frac{6}{10}$ and the probability of playing tennis is $\frac{5}{8}$. What percentage of pupils play both football and tennis?
- (ii) 2 in 5 people in a school study Spanish and 1 in 3 study French. Half of the school study either French or Spanish. What fraction study both French and Spanish?
- (c) (i) 90% of pupils in a school have a Facebook account and 3 out of 5 have a Twitter account. One twentieth of pupils have neither a Facebook account nor a Twitter account. What percentage are on both Facebook and Twitter?
- (ii) 25% of teams in a football league have French players and a third have Italian players. 60% have neither French nor Italian players. What percentage have both French and Italian players?

3. Simplify the following expressions where possible:

- (a) $P(x > 2 \cap x > 4)$
(b) $P(y \leq 3 \cup y < 2)$
(c) $P(a < 3 \cap a > 4)$
(d) $P(a < 5 \cup a \geq 0)$
(e) $P(\text{apple} \cup \text{fruit})$
(f) $P(\text{apple} \cap \text{fruit})$
(g) $P(\text{multiple of } 4 \cap \text{multiple of } 2)$
(h) $P(\text{square} \cup \text{rectangle})$
(i) $P(\text{blue} \cap (\text{blue} \cup \text{red}))$
(j) $P(\text{blue} \cap (\text{blue} \cap \text{red}))$

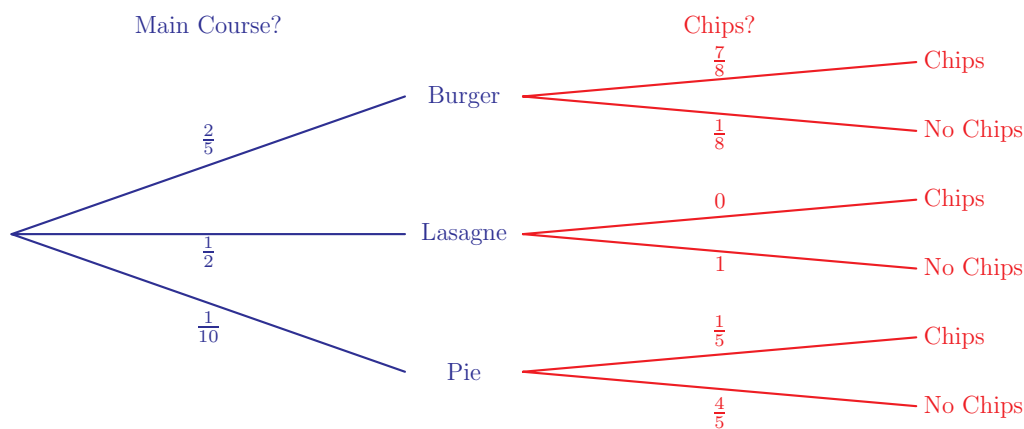
4. In a survey, 60% of people are in favour of building a new primary school and 85% are in favour of building a new library. Half of all those surveyed would like both a new primary school and a new library. What percentage supported neither a new library nor a new primary school? [5 marks]

5. If $P(A) = 0.2$, $P(A \cap B) = 0.1$ and $P(A \cup B) = 0.7$,
find $P(B')$. [5 marks]
6. Events A and B satisfy $P((A \cap B)') = 0.2$, $P(A) = P(B) = 0.5$.
Find $P(A \cap B')$. [5 marks]
7. An integer is chosen at random from the first one thousand
positive integers. Find the probability that the integer
chosen is:
(a) a multiple of 6
(b) a multiple of both 6 and 8. [5 marks]

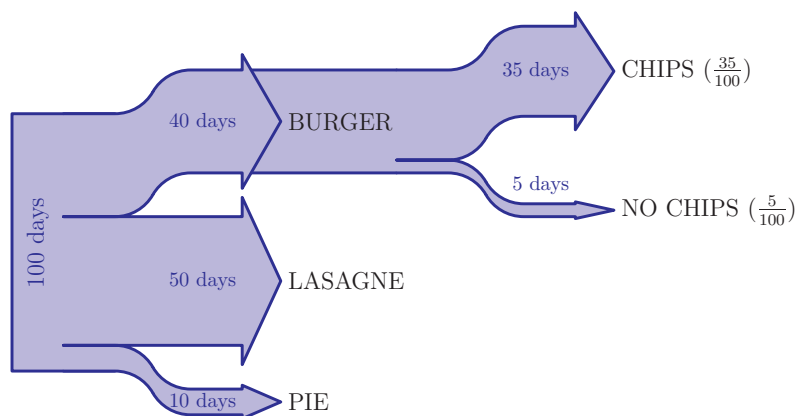
22C Tree diagrams and finding the intersection

Venn diagrams provide a formula linking intersection and union. However they do not help us find a formula for either intersection or union by themselves. To do this it is useful to consider a second method: tree diagrams. When several events happen, either in succession or simultaneously, a **tree diagram** is a way of listing all the possible outcomes. It starts with branches for all the possible outcomes for one of the events, then from each branch we list all possible outcomes for the next event. Along each branch we write the probability of taking that branch.

Tree diagrams have an advantage over the sample space method as they can cope with more than two events, and with outcomes which are not equally likely.



We can consider there to be a filtering process at each branch. Suppose we started with 100 days of school food. On $\frac{2}{5}$ of these days (i.e. 40 days) there will be burgers. On $\frac{7}{8}$ of these burger days there will also be chips. So overall there will be 35 out of 100 days with burgers and chips – or a probability of $\frac{7}{20}$.



To find the probability of travelling along each branch we have found a fraction of a fraction. To do this we need to multiply the two probabilities; but what do these two probabilities represent? In our burger and chips case the $\frac{2}{5}$ represents the probability of burgers, and it might be tempting to say that the $\frac{7}{8}$ is the probability of chips. However, this is not true. Looking at the completed tree diagram we can see that 37 out of 100 days have chips, and this is certainly not $\frac{7}{8}$. The $\frac{7}{8}$ represents the probability of having chips if you already know that there are burgers. This is called **conditional probability**. We use the notation $P(\text{chips} \mid \text{burgers})$, read as ‘the probability of chips given burgers’. This leads to a very important rule.

KEY POINT 22.5

Conditional probability

$$P(A \cap B) = P(B)P(A|B)$$

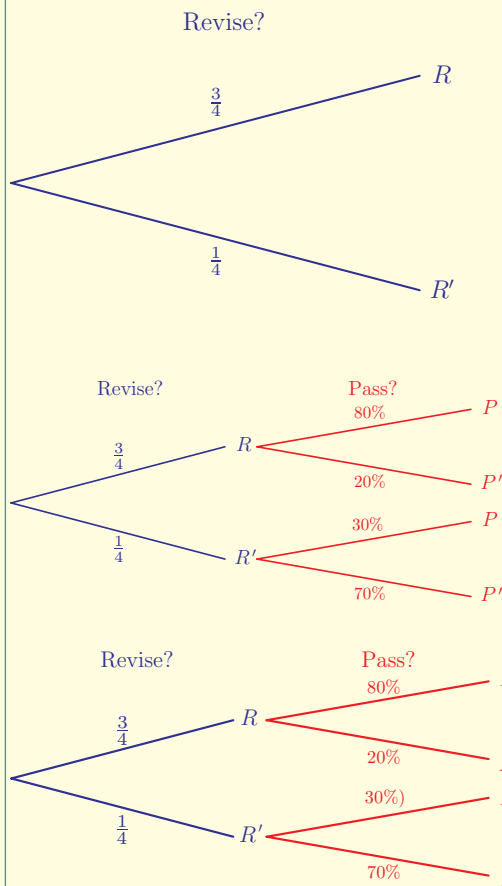
Worked example 22.6

If I revise there is an 80% chance I will pass the maths test, but if I do not revise there is only a 30% chance of passing. I revise for $\frac{3}{4}$ of tests. What proportion of tests do I pass?

Decide which probability (revise or pass) is not conditional. Start the tree diagram with this event. The probability of passing the test is conditional on revision, so the revision branches have to come first

Add the conditional event

Identify which branches result in passing the test. Multiply to find the probability at the end of each branch



$$\begin{aligned}
 P(\text{passing}) &= P(\text{revising} \cap \text{passing}) + P(\text{not revising} \cap \text{passing}) \\
 &= \frac{24}{40} + \frac{3}{40} \\
 &= \frac{27}{40}
 \end{aligned}$$

EXAM HINT

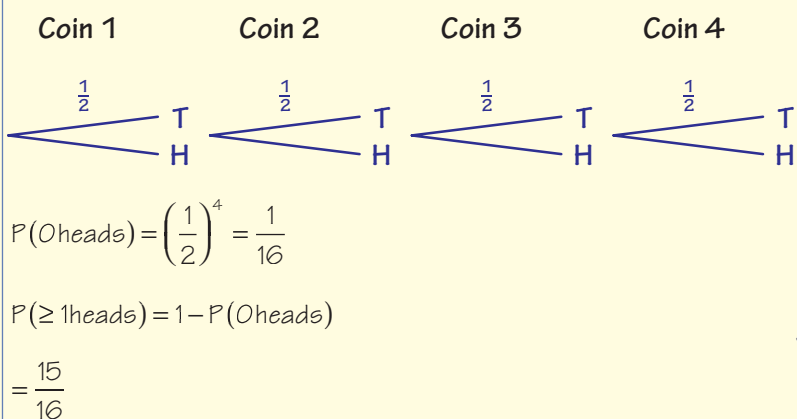
The question in Worked example 22.6 uses the words chance and proportion. These are just other words for probability. Make sure you do not get put off by unusually worded questions.

Sometimes a question may ask for the probability of an event which is quite difficult to find directly, but it is much easier to find the probability of its complement.

Worked example 22.7

Find the probability of getting at least 1 head when you toss four fair coins.

We could draw a tree diagram to find the probability of 1, 2, 3 or 4 heads, but it is easier to find the complement (0 heads)



EXAM HINT

When you are asked to find the probability of 'at least...' or 'at most...' finding the complement is frequently a good idea.

Exercise 22C

1. (a) (i) $P(A) = 0.4$ and $P(B | A) = 0.3$. Find $P(A \cap B)$.
(ii) $P(X) = \frac{3}{5}$ and $P(Y | X) = 0$. Find $P(X \cap Y)$.
- (b) (i) $P(A) = 0.3$, $P(B) = 0.2$ and $P(B | A) = 0.8$.
Find $P(A \cap B)$.

(ii) $P(A) = 0.4$, $P(B) = 0.8$ and $P(A | B) = 0.3$.

Find $P(A \cap B)$.

(c) (i) $P(A) = \frac{2}{5}$, $P(B) = \frac{1}{3}$ and $P(A | B) = \frac{1}{4}$.

Find $P(A \cup B)$.

(ii) $P(A) = \frac{3}{4}$, $P(B) = \frac{1}{4}$ and $P(B | A) = \frac{1}{3}$.

Find $P(A \cup B)$.

2. A class contains 6 boys and 8 girls. Two are picked at random. What is the probability that they are both boys? [4 marks]
3. A bag contains 4 red balls, 3 blue balls and 2 green balls. A ball is chosen at random from the bag and is not replaced. A second ball is chosen. Find the probability of choosing one green ball and one blue ball in any order. [5 marks]
4. Given that $P(X) = \frac{1}{3}$, $P(Y | X) = \frac{2}{9}$ and $P(Y | X') = \frac{1}{3}$, find:
(a) $P(Y')$
(b) $P(X' \cup Y')$ [6 marks]
5. A factory has two machines making widgets. The older machine has larger capacity, so it makes 60% of the widgets, but 6% are rejected by quality control. The newer machine has only a 3% rejection rate. Find the probability that a randomly selected widget is rejected. [5 marks]
6. The school tennis league consists of 12 players. Daniel has a 30% chance of winning any game against a higher-ranked player, and a 70% chance of winning any game against a lower-ranked player. If Daniel is currently in third place, find the probability that he wins his next game against a random opponent. [5 marks]
7. There are 36 disks in a bag. Some of them are black and the rest are white. Two are simultaneously selected at random. Given that the probability of selecting two disks of the same colour is equal to the probability of selecting two disks of different colour, how many black disks are there in the bag? [6 marks]

22D Independent events

We can now evaluate the intersection and union of two events A and B if we know $P(A|B)$ or $P(B|A)$, but finding this can be quite difficult. There is one important exception, when the two events do not affect each other. These are called **independent events**. In this case knowing that B has occurred has no impact on the probability of A occurring, so for independent events $P(A|B) = P(A)$. However, we know that $P(A \cap B) = P(A|B) P(B)$.

KEY POINT 22.6

For independent events: $P(A \cap B) = P(A)P(B)$.

As well as being true for all independent events, Key point 22.6 is actually a defining feature of independent events. So to show that two events are independent we need to show that they satisfy this equation.

All strawberries are red, but not all red things are strawberries. There can be problems distinguishing between a property and a defining feature. A circle has a constant width, but does having a constant width make a plane shape a circle?



Worked example 22.8

$P(A) = 0.5$, $P(B) = 0.2$ and $P(A \cup B) = 0.6$. Are the events A and B independent?

Use the information to find $P(A \cap B)$

$$\begin{aligned} P(A \cup B) &= P(A) + P(B) - P(A \cap B) \\ \Rightarrow 0.6 &= 0.5 + 0.2 - P(A \cap B) \\ \Rightarrow P(A \cap B) &= 0.1 \end{aligned}$$

Evaluate $P(A)P(B)$

$$P(A)P(B) = 0.1$$

Compare the two

$$\therefore P(A \cap B) = P(A)P(B)$$

So A and B are independent

If we know that two events are independent we can use this to help calculate other probabilities.

Worked example 22.9

A and B are independent events with $P(A \cup B) = 0.8$ and $P(A) = 0.2$. Find $P(B)$.

Write $P(A \cap B)$ in terms of other probabilities

Use independence

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$\Rightarrow 0.8 = 0.2 + P(B) - P(A \cap B)$$

$$= 0.2 + P(B) - 0.2 \times P(B)$$

$$\Rightarrow 0.6 = 0.8 \times P(B)$$

$$\Rightarrow P(B) = \frac{3}{4} \text{ (or } 0.75\text{)}$$

Exercise 22D

1. Events A and B are independent.

(a) (i) $P(A) = 0.3$ and $P(B) = 0.7$. Find $P(A \cap B)$.

(ii) $P(A) = \frac{1}{5}$ and $P(B) = \frac{1}{3}$. Find $P(A \cap B)$.

(b) (i) $P(A) = \frac{4}{5}$ and $P(A \cap B) = \frac{3}{7}$. Find $P(B)$.

(ii) $P(A \cap B) = 0.5$ and $P(B) = 0.9$. Find $P(A)$.

(c) (i) $P(A) = 40\%$ and $P(B) = 16\%$. Find $P(A \cup B)$.

(ii) $P(A) = 0.2$ and $P(B) = \frac{1}{4}$. Find $P(A \cup B)$.

(d) (i) $P(A \cup B) = 0.6$ and $P(A) = 0.4$. Find $P(B)$.

(ii) $P(A \cup B) = 0.5$ and $P(A) = 0.1$. Find $P(B)$.

2. Determine which of the following pairs of events are independent:

(a) (i) $P(A) = 0.6$, $P(B) = 0.5$ and $P(A \cap B) = 0.3$

(ii) $P(A) = 0.8$, $P(B) = 0.1$ and $P(A \cap B) = 0.05$

(b) (i) $P(A) = \frac{1}{5}$, $P(B) = \frac{1}{4}$ and $P(A \cup B) = \frac{2}{5}$

(ii) $P(A) = 56\%$, $P(B) = 32\%$ and $P(A \cup B) = 72\%$

3. The independent events A and B are such that $P(A) = 0.6$ and $P(A \cup B) = 0.72$. Find:
- $P(B)$
 - the probability that either A occurs or B occurs, but not both. [6 marks]
4. A school has two photocopiers, one for teachers and one for pupils. The probability of the teachers' photocopier working is 92%. The probability of the students' photocopier working is 68%. The two outcomes do not affect each other. What is the probability that:
- both photocopiers are working
 - neither photocopier is working
 - at least one photocopier is working. [7 marks]
5. As part of a promotion a toy is put in each packet of crisps sold. There are eight different toys available. Each toy is equally likely to be found in any packet of crisps.
- David buys four packets of crisps.
- Find the probability that the four toys in these packets are all different.
 - Of the eight toys in the packets, his favourites are the yo-yo and the gyroscope. Find the probability that he finds at least one of his favourite toys in these four packets. [7 marks]
6. Given that events A and B are independent with $P(A \cap B) = 0.3$ and $P(A \cap B') = 0.6$, find $P(A \cup B)$. [5 marks]

22E Counting principles in probability

Counting principles can be used to calculate probabilities in some problems that would be extremely difficult in any other way. A random arrangement of a collection of objects forms a sample space so we can find the probability of particular condition A occurring.

KEY POINT 22.7

$$P(A) = \frac{\text{number of ways in which } A \text{ occurs}}{\text{total number of ways}}$$

In chapter 1 you learnt how to count the number of ways in which events can occur.